Thiol S-methyltransferase TMT1B structure report

METTL7B / UniProt Q6UX53 / Structure 2

PURPOSE

This report describes the selected computational pocket, the residues that define it, and the ligand-contact evidence that supports or extends its boundary.

CHOSEN POCKET	RESIDUES	VOLUME	DRUGGABILITY
FPOCKET 2	47	1690.538 A3	0.832

EXECUTIVE INTERPRETATION

What the evidence says

The chosen site is FPOCKET 2 with 47 residues. SAM has 16 strong contacts within 4.0 Angstrom and 9 near contacts between 4.0 and 4.5 Angstrom. 3 direct contacts lie outside the chosen pocket. SAH has 17 strong contacts within 4.0 Angstrom and 7 near contacts between 4.0 and 4.5 Angstrom. 2 direct contacts lie outside the chosen pocket. The original pocket membership is unchanged.

Across all ligands, A:125 ALA (A), A:100 ASN (N) consistently contact the predicted ligand outside the chosen fpocket. Ligand-specific boundary contacts are A:76 GLU (E).

BOUNDARY RULE

The original fpocket membership is preserved. BioHub contacts outside fpocket are reported as evidence, not silently added to the pocket.

Generated from current database state on Jul 29, 2026 20:52 PDT.

POCKET DEFINITION

The residues that define the chosen pocket

FPOCKET 2 contains 47 ordered residues. The list below uses chain, residue number, three-letter name, and one-letter sequence code so every entry can be traced back to the structure.

SCORE	DRUGGABILITY	VOLUME
0.003	0.832	1690.538 A3

A:29 GLN (Q)   A:33 LYS (K)   A:36 PHE (F)   A:39 LEU (L)   A:40 MET (M)   A:43 LEU (L)   A:44 THR (T)   A:47 SER (S)   A:48 ASN (N)   A:55 LYS (K)   A:77 LEU (L)   A:78 GLY (G)   A:79 CYS (C)   A:80 GLY (G)   A:81 THR (T)   A:83 ALA (A)   A:84 ASN (N)   A:97 LEU (L)   A:98 ASP (D)   A:99 PRO (P)   A:103 PHE (F)   A:126 PRO (P)   A:127 GLY (G)   A:128 GLU (E)   A:144 THR (T)   A:145 LEU (L)   A:146 VAL (V)   A:148 CYS (C)   A:149 SER (S)   A:150 VAL (V)   A:151 GLN (Q)   A:152 SER (S)   A:155 LYS (K)   A:173 TRP (W)   A:175 HIS (H)   A:195 TRP (W)   A:198 ILE (I)   A:199 GLY (G)   A:200 ASP (D)   A:201 GLY (G)   A:202 CYS (C)   A:228 PRO (P)   A:229 LEU (L)   A:231 TRP (W)   A:232 LEU (L)   A:234 VAL (V)   A:237 HIS (H)

LIGAND-CONDITIONED EVIDENCE

SAM and the pocket boundary

SAM places 25 residues within the 4.5 A direct-contact boundary. 16 are strong contacts within 4.0 A, and 9 are near contacts between 4.0 and 4.5 A. 22 direct contacts support the chosen fpocket, while 3 direct contacts fall outside its stored membership. The interface pTM is 0.984.

STRONG <= 4.0 A	4.0-4.5 A	DIRECT CONTACTS	OUTSIDE FPOCKET
16	9	25	3

CONTACTS OUTSIDE FPOCKET

A:76 GLU (E76) is 4.4 A from the ligand and is classified as near. A:100 ASN (N100) is 3.9 A from the ligand and is classified as strong. A:125 ALA (A125) is 4.5 A from the ligand and is classified as near.

RESIDUE	SEQUENCE	DISTANCE	CLASS	FPOCKET
A:36 PHE	F36	4.01 A	NEAR	Yes
A:40 MET	M40	3.52 A	STRONG	Yes
A:44 THR	T44	3.49 A	STRONG	Yes
A:76 GLU	E76	4.42 A	NEAR	No
A:77 LEU	L77	4.34 A	NEAR	Yes
A:78 GLY	G78	2.76 A	STRONG	Yes
A:79 CYS	C79	3.81 A	STRONG	Yes
A:80 GLY	G80	3.53 A	STRONG	Yes
A:83 ALA	A83	4.46 A	NEAR	Yes
A:84 ASN	N84	4.46 A	NEAR	Yes
A:97 LEU	L97	4.31 A	NEAR	Yes
A:98 ASP	D98	2.49 A	STRONG	Yes
A:99 PRO	P99	3.23 A	STRONG	Yes
A:100 ASN	N100	3.89 A	STRONG	No
A:103 PHE	F103	3.16 A	STRONG	Yes
A:125 ALA	A125	4.47 A	NEAR	No
A:126 PRO	P126	3.65 A	STRONG	Yes
A:127 GLY	G127	2.81 A	STRONG	Yes
A:128 GLU	E128	2.89 A	STRONG	Yes
A:144 THR	T144	2.77 A	STRONG	Yes
A:145 LEU	L145	3.14 A	STRONG	Yes
A:146 VAL	V146	3.83 A	STRONG	Yes
A:149 SER	S149	4.18 A	NEAR	Yes
A:150 VAL	V150	3.51 A	STRONG	Yes
A:200 ASP	D200	4.00 A	NEAR	Yes

LIGAND-CONDITIONED EVIDENCE

SAH and the pocket boundary

SAH places 24 residues within the 4.5 A direct-contact boundary. 17 are strong contacts within 4.0 A, and 7 are near contacts between 4.0 and 4.5 A. 22 direct contacts support the chosen fpocket, while 2 direct contacts fall outside its stored membership. The interface pTM is 0.985.

STRONG <= 4.0 A	4.0-4.5 A	DIRECT CONTACTS	OUTSIDE FPOCKET
17	7	24	2

CONTACTS OUTSIDE FPOCKET

A:100 ASN (N100) is 3.9 A from the ligand and is classified as strong. A:125 ALA (A125) is 4.5 A from the ligand and is classified as near.

RESIDUE	SEQUENCE	DISTANCE	CLASS	FPOCKET
A:36 PHE	F36	3.98 A	STRONG	Yes
A:40 MET	M40	3.71 A	STRONG	Yes
A:44 THR	T44	3.45 A	STRONG	Yes
A:77 LEU	L77	4.40 A	NEAR	Yes
A:78 GLY	G78	2.75 A	STRONG	Yes
A:79 CYS	C79	3.82 A	STRONG	Yes
A:80 GLY	G80	3.53 A	STRONG	Yes
A:83 ALA	A83	4.43 A	NEAR	Yes
A:97 LEU	L97	4.33 A	NEAR	Yes
A:98 ASP	D98	2.46 A	STRONG	Yes
A:99 PRO	P99	3.23 A	STRONG	Yes
A:100 ASN	N100	3.92 A	STRONG	No
A:103 PHE	F103	3.17 A	STRONG	Yes
A:125 ALA	A125	4.47 A	NEAR	No
A:126 PRO	P126	3.66 A	STRONG	Yes
A:127 GLY	G127	2.82 A	STRONG	Yes
A:128 GLU	E128	2.91 A	STRONG	Yes
A:144 THR	T144	2.72 A	STRONG	Yes
A:145 LEU	L145	3.17 A	STRONG	Yes
A:146 VAL	V146	3.82 A	STRONG	Yes
A:149 SER	S149	4.10 A	NEAR	Yes
A:150 VAL	V150	3.53 A	STRONG	Yes
A:151 GLN	Q151	4.49 A	NEAR	Yes
A:200 ASP	D200	4.17 A	NEAR	Yes

EVIDENCE APPENDIX

Complete ligand proximity records

Strong contacts are within 4.0 A. Near contacts are greater than 4.0 A and within 4.5 A. Context residues are greater than 4.5 A and within 6.0 A. Distances are minimum heavy-atom distances in the BioHub predicted protein-ligand complex.

SAM

SAM complete residue evidence

RESIDUE	SEQUENCE	DISTANCE	CLASS	FPOCKET
A:33 LYS	K33	5.44 A	CONTEXT	Yes
A:36 PHE	F36	4.01 A	NEAR	Yes
A:40 MET	M40	3.52 A	STRONG	Yes
A:44 THR	T44	3.49 A	STRONG	Yes
A:55 LYS	K55	4.77 A	CONTEXT	Yes
A:76 GLU	E76	4.42 A	NEAR	No
A:77 LEU	L77	4.34 A	NEAR	Yes
A:78 GLY	G78	2.76 A	STRONG	Yes
A:79 CYS	C79	3.81 A	STRONG	Yes
A:80 GLY	G80	3.53 A	STRONG	Yes
A:81 THR	T81	4.98 A	CONTEXT	Yes
A:82 GLY	G82	5.90 A	CONTEXT	No
A:83 ALA	A83	4.46 A	NEAR	Yes
A:84 ASN	N84	4.46 A	NEAR	Yes
A:97 LEU	L97	4.31 A	NEAR	Yes
A:98 ASP	D98	2.49 A	STRONG	Yes
A:99 PRO	P99	3.23 A	STRONG	Yes
A:100 ASN	N100	3.89 A	STRONG	No
A:103 PHE	F103	3.16 A	STRONG	Yes
A:125 ALA	A125	4.47 A	NEAR	No
A:126 PRO	P126	3.65 A	STRONG	Yes
A:127 GLY	G127	2.81 A	STRONG	Yes
A:128 GLU	E128	2.89 A	STRONG	Yes
A:143 CYS	C143	5.71 A	CONTEXT	No
A:144 THR	T144	2.77 A	STRONG	Yes
A:145 LEU	L145	3.14 A	STRONG	Yes
A:146 VAL	V146	3.83 A	STRONG	Yes
A:148 CYS	C148	5.46 A	CONTEXT	Yes
A:149 SER	S149	4.18 A	NEAR	Yes
A:150 VAL	V150	3.51 A	STRONG	Yes
A:151 GLN	Q151	4.62 A	CONTEXT	Yes
A:156 VAL	V156	5.22 A	CONTEXT	No
A:200 ASP	D200	4.00 A	NEAR	Yes

SAH

SAH complete residue evidence

RESIDUE	SEQUENCE	DISTANCE	CLASS	FPOCKET
A:33 LYS	K33	5.65 A	CONTEXT	Yes
A:36 PHE	F36	3.98 A	STRONG	Yes
A:40 MET	M40	3.71 A	STRONG	Yes
A:44 THR	T44	3.45 A	STRONG	Yes
A:55 LYS	K55	4.83 A	CONTEXT	Yes
A:76 GLU	E76	4.51 A	CONTEXT	No
A:77 LEU	L77	4.40 A	NEAR	Yes
A:78 GLY	G78	2.75 A	STRONG	Yes
A:79 CYS	C79	3.82 A	STRONG	Yes
A:80 GLY	G80	3.53 A	STRONG	Yes
A:81 THR	T81	5.01 A	CONTEXT	Yes
A:82 GLY	G82	5.89 A	CONTEXT	No
A:83 ALA	A83	4.43 A	NEAR	Yes
A:84 ASN	N84	4.50 A	CONTEXT	Yes
A:97 LEU	L97	4.33 A	NEAR	Yes
A:98 ASP	D98	2.46 A	STRONG	Yes
A:99 PRO	P99	3.23 A	STRONG	Yes
A:100 ASN	N100	3.92 A	STRONG	No
A:103 PHE	F103	3.17 A	STRONG	Yes
A:125 ALA	A125	4.47 A	NEAR	No
A:126 PRO	P126	3.66 A	STRONG	Yes
A:127 GLY	G127	2.82 A	STRONG	Yes
A:128 GLU	E128	2.91 A	STRONG	Yes
A:143 CYS	C143	5.73 A	CONTEXT	No
A:144 THR	T144	2.72 A	STRONG	Yes
A:145 LEU	L145	3.17 A	STRONG	Yes
A:146 VAL	V146	3.82 A	STRONG	Yes
A:148 CYS	C148	5.83 A	CONTEXT	Yes
A:149 SER	S149	4.10 A	NEAR	Yes
A:150 VAL	V150	3.53 A	STRONG	Yes
A:151 GLN	Q151	4.49 A	NEAR	Yes
A:156 VAL	V156	5.23 A	CONTEXT	No
A:200 ASP	D200	4.17 A	NEAR	Yes